

Conservative Euler–BIE/VOF coupling for overturning free-surface waves over variable bathymetry

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Abstract

A hybrid method is developed for two-dimensional overturning waves over variable bathymetry. A fully nonlinear Euler boundary-integral equation (BIE) advances the irrotational pre-contact flow, while an embedded two-phase volume-of-fluid (VOF) solver provides the continuation when the interface approaches self-contact. The BIE combines Kress product quadrature with a nonlocal-gap trigger for oversampled close evaluation and represents the adaptive boundary state by integrated panel circulations. At transfer, shared face-averaged BIE fluxes are projected onto a vertex streamfunction; the resulting MAC face field is divergence-free by construction. The liquid aperture $q = fc_s$ is then transported with a shared embedded-boundary flux, including an exact pure-state rule and a capacity restriction confined to the mixed-interface band.

Verification separates operator accuracy, conservation and interface geometry. On a manufactured folded boundary, oversampling reduces the relative Dirichlet-to-Neumann error from 6.36 to 6.91×10^{-2} at 64 panels despite similarly small linear residuals. A solitary-wave translation has shape and speed errors 1.82×10^{-5} and 8.40×10^{-5} . Manufactured aperture transport is first-order on the three finest grids and preserves liquid volume and $0 \leq q \leq c_s$ to roundoff. The constrained handoff has 1.68% face-flux error, 3.12% surface velocity error and 1.07×10^{-14} face-divergence RMS.

The coupled breaker is continued to nondimensional time 6 on three nested receiver grids. Maximum relative liquid-volume drift is 8.56×10^{-8} , 6.43×10^{-8} and 5.67×10^{-9} on levels 8–10, with no aperture-bound violation. Level 10 produces a transient disconnected gas component at $t = 1.96$, but the event is absent at the two coarser levels and its detected onset varies by 0.06 under conservative raster coarsening. At $t = 6$, the level-9/10 upper-profile and crest differences are 8.18% and 6.25%. The method therefore achieves stable conservative continuation and exposes a resolution-dependent topology transition, but the present calculation does not establish grid-converged wave impact or air entrainment.

1 Introduction

Boundary-integral descriptions of inviscid water waves resolve vertical and overturning free surfaces without a height-function restriction [1, 4, 11]. They remain efficient while the liquid is irrotational and the interface is a single non-self-intersecting curve, but cease to describe entrained air, vorticity production and topology change after impact. Fully nonlinear BEM calculations have reproduced solitary-wave shoaling and classified breaking regimes over beaches and obstacles [6], while two-phase VOF methods naturally continue through interface reconnection.

Hybridisation alone is not new. Khait and Ma coupled a quasi-potential BEM to a two-phase incompressible VOF solver for breaking wave trains [10]. Kanehira et al. recently embedded a GPU-SPH region in a three-dimensional fully nonlinear potential-flow calculation and compared jet and cavity dynamics with experiments [9]. The numerical question considered here is narrower: can an adaptive circulation-carrying BIE state be transferred to an embedded two-phase receiver while conserving the appropriate liquid measure and satisfying the receiver’s discrete divergence and momentum constraints?

The method has four coupled elements. First, a nonlocal geometric indicator activates oversampled quadrature when two parameter-distant surface branches approach. Second, integrated

panel circulation is preserved under adaptive redistribution. Third, face-averaged velocity rather than independent cell-centred samples is transferred to a divergence-conforming MAC field. Fourth, the full-cell liquid aperture $q = fc_s$ is transported by a shared VOF/embedded-boundary flux. The calculations below test these components and then apply the complete coupling to an overturning breaker on three receiver grids.

2 Mathematical formulation

2.1 Pre-contact free-surface flow

Let $\Omega(t) \subset \mathbb{R}^2$ be the liquid domain. With velocity $\mathbf{u} = \nabla\phi$, the potential satisfies

$$\nabla^2\phi = 0 \quad \text{in } \Omega(t), \quad \partial_n\phi = 0 \quad \text{on the bed } \Gamma_b. \quad (1)$$

The periodic free surface is $\mathbf{X}(\alpha, t) = (x(\alpha, t), z(\alpha, t))$ and carries the trace $\psi = \phi|_{\Gamma_f}$. With normal velocity $V_n = \partial_n\phi$, tangential fluid velocity $V_t = \partial_s\psi$ and freely chosen marker velocity W_t , the surface equations are

$$\mathbf{X}_t = V_n \mathbf{n} + W_t \mathbf{t}, \quad (2)$$

$$\psi_t = \frac{1}{2}V_n^2 - \frac{1}{2}V_t^2 + W_t V_t - gz. \quad (3)$$

For prescribed ψ_f and zero bed flux, Green's identity gives the mixed system

$$[S_{:f} \quad -(\frac{1}{2}I + K)_{:b}] \begin{bmatrix} q_f \\ \phi_b \end{bmatrix} = (\frac{1}{2}I + K)_{:f} \psi_f, \quad (4)$$

where $q_f = \partial_n\phi|_{\Gamma_f}$ and S and K are the periodic single- and double-layer operators. Their Green function is

$$G_L(\Delta x, \Delta z) = -\frac{1}{4\pi} \log \left[2 \left(\cosh \frac{2\pi\Delta z}{L} - \cos \frac{2\pi\Delta x}{L} \right) \right]. \quad (5)$$

Kress product weights integrate the logarithmic self interaction. The incident-wave normal velocity is projected onto the actual bathymetry by subtracting its mean boundary flux and solving the compatible Neumann problem in a zero-mean gauge.

The ordinary self rule becomes under-resolved when two distant arclength segments approach. Define

$$d_{\text{nl}} = \min_{d_s(i,j) > 8\overline{\Delta s}} \|\mathbf{X}_i - \mathbf{X}_j\|, \quad \overline{\Delta s} = \frac{2\pi}{N} |\overline{\mathbf{X}_\alpha}|. \quad (6)$$

When $d_{\text{nl}}/\overline{\Delta s} < 1.5$, the complete free-surface self block is reevaluated after trigonometric interpolation to pN points, with

$$p = \text{clip}_{[2,16]} \lceil 4\overline{\Delta s}/d_{\text{nl}} \rceil. \quad (7)$$

The density and geometry are oversampled together so that the Kress splitting remains consistent.

2.2 Adaptive circulation representation

The surface potential trace is equivalently represented by sheet strength $\gamma = -\partial_s\psi$. For a panel j the integrated circulation

$$\Gamma_j = - \int_{s_j}^{s_{j+1}} \gamma \, ds = \psi_{j+1} - \psi_j \quad (8)$$

is used as the conservative carrier during marker redistribution. New panel endpoints are chosen from a smoothed monitor depending on curvature, sheet strength and inverse nonlocal gap; the cumulative circulation is interpolated monotonically and then differenced on the new partition. This preserves total circulation to roundoff. These boundary circulations represent the irrotational potential trace; they are not bulk material vortices.

2.3 Conservative embedded liquid aperture

For a Cartesian receiver cell K_i , define the full-cell physical liquid aperture

$$q_i = \frac{|K_i \cap \Omega_\ell \cap \Omega_f|}{|K_i|}, \quad 0 \leq q_i \leq c_{s,i}, \quad (9)$$

where c_s is the open-fluid volume fraction and $f = q/c_s$ is conditional on the open part of the cell. This full-cell measure is consistent with solid-aware VOF/embedded-boundary formulations and conservative contact-line treatments [3, 7, 8, 14]. One oriented swept volume is assigned to every shared face,

$$F_e = |S_e(\Delta t) \cap \Omega_\ell \cap \Omega_f|, \quad q_i^{n+1} = q_i^n - |K_i|^{-1} \sum_{e \in \partial K_i} \text{sgn}(i, e) F_e. \quad (10)$$

Because the same F_e enters adjacent cells with opposite signs, global liquid volume telescopes exactly. A pure donor satisfies $F_{\ell,e} = 0$ for $f_d = 0$ and $F_{\ell,e} = F_{f,e}$ for $f_d = 1$; only genuinely mixed cut cells require a capacity restriction. The restriction is applied to mixed cells and their face-connected one-cell halo,

$$\Delta t \leq 0.45 \min_{i \in \mathcal{I}} \frac{c_{s,i} \Delta_i}{\max_{e \in \partial K_i} |u_e f_{s,e}|}. \quad (11)$$

2.4 Divergence-conforming state transfer

The target liquid momentum is evaluated from boundary data,

$$\mathbf{P}^* = \rho \int_{\Omega} \nabla \phi \, dA = \rho \oint_{\partial \Omega} \phi \mathbf{n} \, ds. \quad (12)$$

Two-point Gauss quadrature supplies the BIE velocity averaged over every receiver face. A periodic vertex streamfunction χ then defines

$$u_{i-1/2,k} = \frac{\chi_{i-1/2,k+1/2} - \chi_{i-1/2,k-1/2}}{\Delta z}, \quad w_{i,k-1/2} = -\frac{\chi_{i+1/2,k-1/2} - \chi_{i-1/2,k-1/2}}{\Delta x}. \quad (13)$$

The standard face-difference divergence of this field vanishes identically. Compatibility projections and divergence-preserving transfer between non-conforming or multilevel grids provide the relevant numerical context [2, 5, 13]. The sparse fit minimises

$$\|W_f(C_h \chi - \bar{\mathbf{u}})\|_2^2 + \lambda_s \|S\chi - \mathbf{u}_s\|_2^2 + \lambda_b \|B\chi\|_2^2 + \lambda_P \|M\chi - \mathbf{P}^*/\rho\|_2^2 + \lambda_e \|E\chi\|_2^2 + \lambda_h \|L_h \chi\|_2^2. \quad (14)$$

Here $B\chi = 0$ makes the bed a streamline, M imposes momentum, E suppresses exterior kinetic energy and L_h regularises unconstrained modes. After AMR, the face flux transported by the mesh is retained and a receiver-grid Helmholtz correction restores compatibility; no global velocity shift is applied.

3 Verification

The tests are grouped by the property they measure. Analytic and manufactured solutions test the BIE and aperture operators; algebraic residuals test the handoff constraints; long coupled runs test conservation and boundedness. Interface contact is assessed separately in Section 4.

For a flat surface Fourier mode, the Dirichlet-to-Neumann eigenvalue and the discrete velocity agree to machine precision. A fully nonlinear solitary wave propagated over one reference interval has relative shape error 1.82×10^{-5} and speed error 8.40×10^{-5} . On a manufactured folded interface, the unoversampled 64-panel discretisation has a relative DNO error of 6.36 even though its linear residual is near 10^{-13} ; order-16 oversampling reduces the error to 6.91×10^{-2} (figure 1).

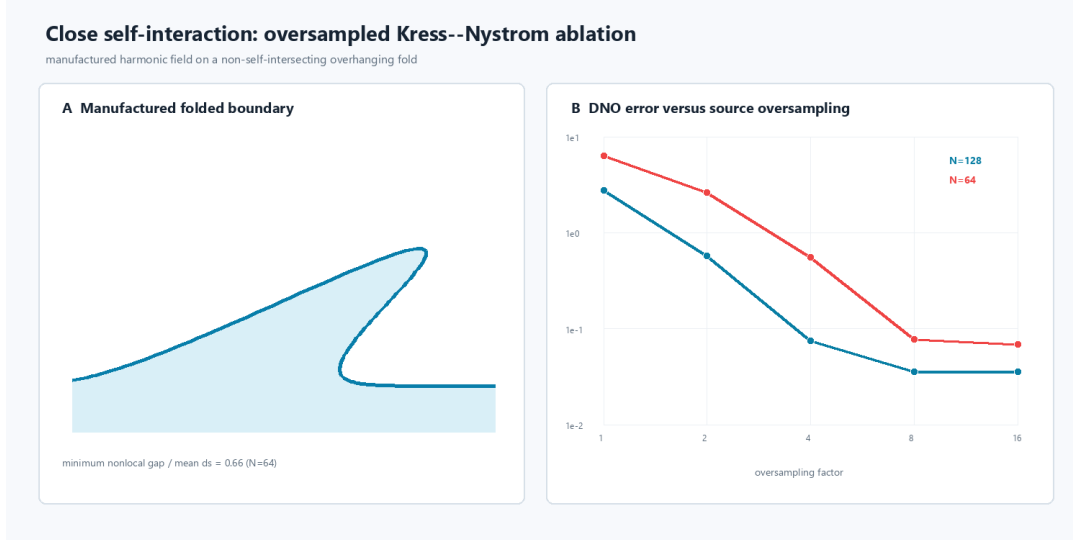


Figure 1: Close-interaction quadrature on a manufactured folded interface. The boundary-integral residual alone does not diagnose the unresolved parameter-distant branch; geometric oversampling reduces the DNO error.

For sinusoidal-interface translation, the relative $L^1(q)$ errors on levels 6–8 are 0.01378, 0.007689 and 0.003538, corresponding to observed orders 0.84 and 1.12. Fixed, frozen-AMR and every-step-AMR volume drift remains below 2.0×10^{-13} . An inclined-wall test gives errors 0.02325, 0.01247 and 0.006154 on levels 7–9 (orders 0.90 and 1.02) while preserving volume and bounds to roundoff. Restricting the cut-cell timestep to the interface band requires 211 steps at level 7, compared with 758 when every cut cell is included.

Independent cell-centred samples of the late BIE velocity have a 15.92% best-fit error in the range of the receiver curl. Shared face averages followed by equation (14) reduce the face-flux error to 1.6835% and the surface-velocity error to 3.1229%. Relative momentum error is 1.99×10^{-9} and the finite-volume divergence RMS is 1.07×10^{-14} . GPU sparse LSQR takes 12.76 s versus 69.54 s for the CPU solve and changes the trusted-face field by only 4.58×10^{-8} relative (figure 2). This acceleration applies to the sparse handoff fit; the two-phase evolution reported below is CPU-serial.

4 Three-grid overturning-wave experiment

4.1 Configuration and event definition

The source is an active $N = 576$ BIE state over a periodic 1:30 beach and return slope, transferred at nondimensional BIE time 20.303125. Conservative raster overlap changes volume by 2.45×10^{-13} relative to the source; reconciliation with the analytic embedded bed changes 0.0868%. Receiver maximum levels 8, 9 and 10 use minimum levels 6, 7 and 8, respectively, and adapt after every accepted physical step. VOF and vorticity fields are sampled every 0.02 through continuation time 6.

A gas component is classified as a resolved closure if it is disconnected from the exterior, does not touch a one-finest-cell bed guard, has area at least Δ^2 , and persists for three output frames. Four-neighbour components are joined across the horizontal periodic seam. Diagnostic coarsening uses exact block liquid-area fractions at scale factors two and three and thresholds 0.25, 0.5 and 0.75. The accepted onset-time variation is at most 0.04.

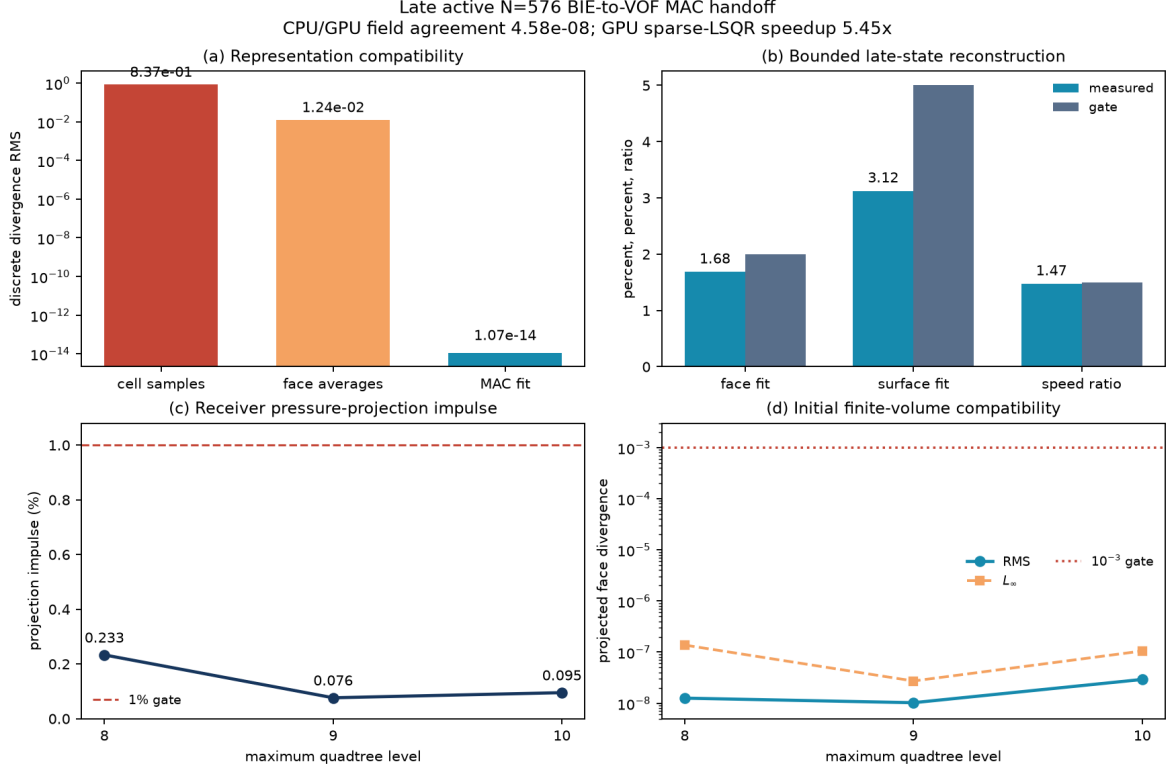


Figure 2: Compatible transfer of the late BIE state. Face averaging reduces the initial representation mismatch and the streamfunction curl enforces finite-volume divergence compatibility. Receiver mass, momentum and post-projection divergence are shown for levels 8–10.

Table 1: Three-grid coupled continuation through $t = 6$. $A_{\max}$ is the largest exterior-disconnected, bed-detached gas area before the one-cell area filter.

Level	$\max \delta V/V $	$\max D_h \mathbf{u} _\infty$	$\max \delta P/P $	$A_{\max}/\Delta^2$	resolved onset
8	8.56×10^{-8}	3.59×10^{-10}	8.24×10^{-5}	0	none
9	6.43×10^{-8}	2.32×10^{-10}	4.04×10^{-5}	0.498	none
10	5.67×10^{-9}	3.86×10^{-9}	6.40×10^{-5}	10.57	1.96

4.2 Conservation and topology

All three continuations remain bounded to $t = 6$. Table 1 summarises the invariant and compatibility histories. The aperture bounds hold to roundoff. The largest post-AMR divergence and relative momentum and kinetic-energy corrections remain below 3.86×10^{-9} , 8.24×10^{-5} and 9.31×10^{-6} , respectively.

Level 8 has no disconnected gas component. Level 9 develops a late component whose maximum area is only $0.498\Delta_9^2$. Level 10 develops a component larger than ten finest-cell areas near $t = 1.96$, but it has disappeared by $t = 2.08$. Across the seven conservative raster probes its onset ranges from 1.90 to 1.96, exceeding the 0.04 tolerance. A later group of cavities at $t = 5.10$ is likewise absent on the coarser grids. Thus the algebraic conservation properties converge more strongly than the observed topology (figures 3 and 4).

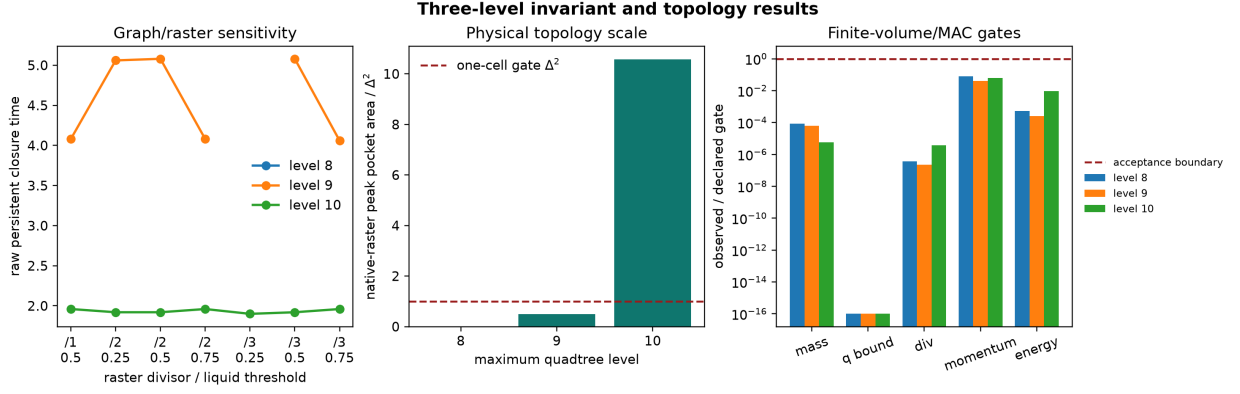


Figure 3: Raster sensitivity, gas-component scale and normalized conservation and MAC compatibility measures. The level-10 component is physically larger than one finest cell, but its time and existence are not reproduced across the receiver grids.

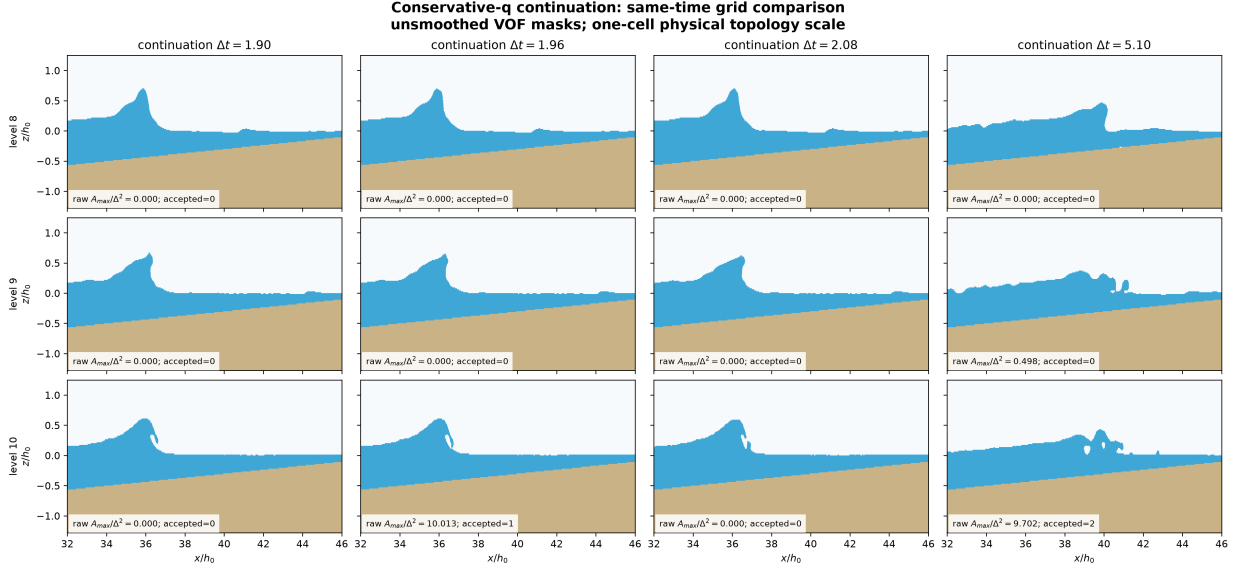


Figure 4: Unsmoothed VOF masks at matched continuation times. The transient level-10 cavity at $t = 1.96$ and the later cavities at $t = 5.10$ are not same-time features of levels 8 and 9. Labels report the largest raw component in units of the local finest-cell area and the number passing the physical area filter.

4.3 Geometric refinement

Upper-surface profiles are extracted at identical physical times without interpolation between frames. Relative RMS differences are normalised by the initial level-10 amplitude. The maximum L8/L9 and L9/L10 profile differences over $0 \leq t \leq 6$ are 11.39% and 8.26%; at $t = 6$ they are 8.12% and 8.18%. The final L9/L10 crest difference is 6.25%, with a maximum of 12.5%. Although the instantaneous Richardson indicator is positive for 80.4% of the last time unit, the fine-pair profile and crest differences both exceed the 5% tolerance (figure 5).

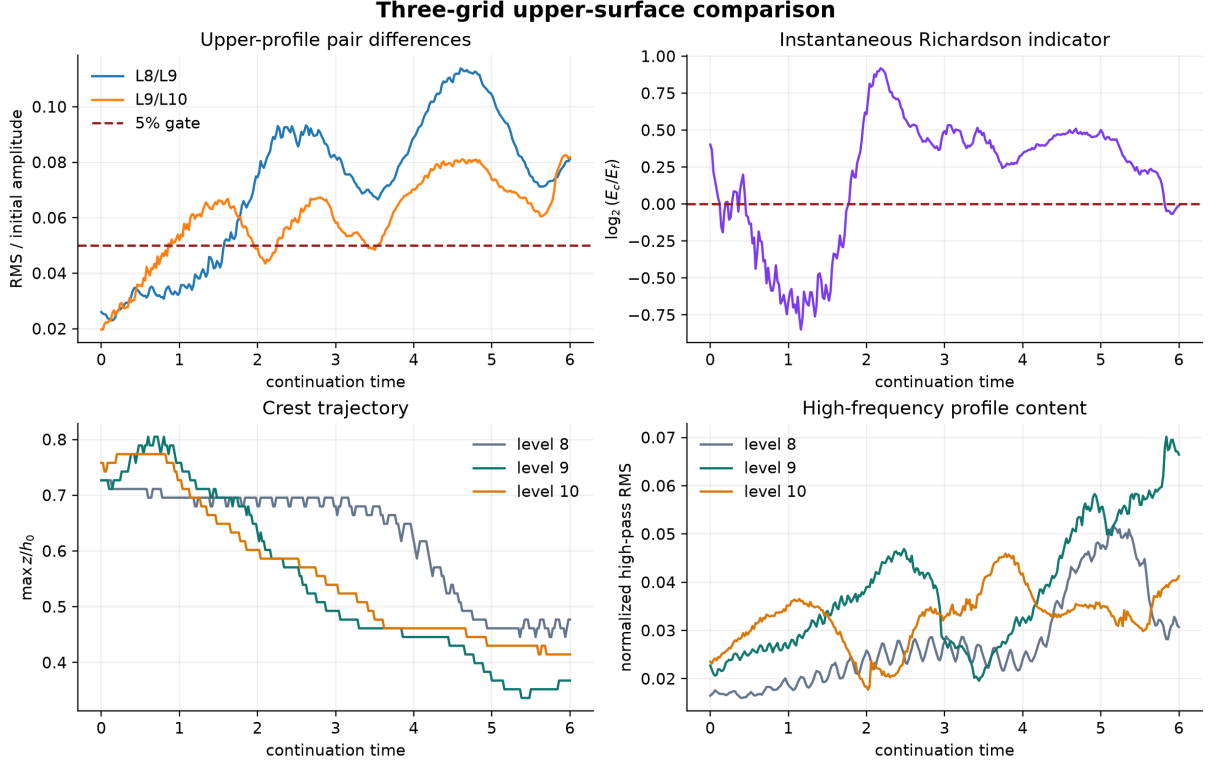


Figure 5: Three-grid upper-surface comparison. Pairwise profile errors remain above the 5% tolerance for much of the continuation, and the final fine-pair profile and crest differences are 8.18% and 6.25%.

The mismatch is not captured by smooth-profile measures alone. On level 9, adapting every second or fourth step changes the upper-profile RMS by at most 3.55% but shifts the first raw subcell component from $t = 4.08$ to about 5.1. Topology is therefore more sensitive to mesh evolution than the bulk profile.

The level-10 vorticity field contains a concentrated signed layer around the curling crest and beneath the emerging cavities (figure 6), consistent with the localisation reported in small-viscosity plunging-wave calculations [12]. The plotted markers sample the Eulerian vorticity quadrature $\Gamma_{ij} \simeq \omega_{ij} \Delta A$; they are not Lagrangian particles. Quantitative post-contact circulation requires a separate refinement study because potential-flow sheet labels cannot be continued through reconnection.

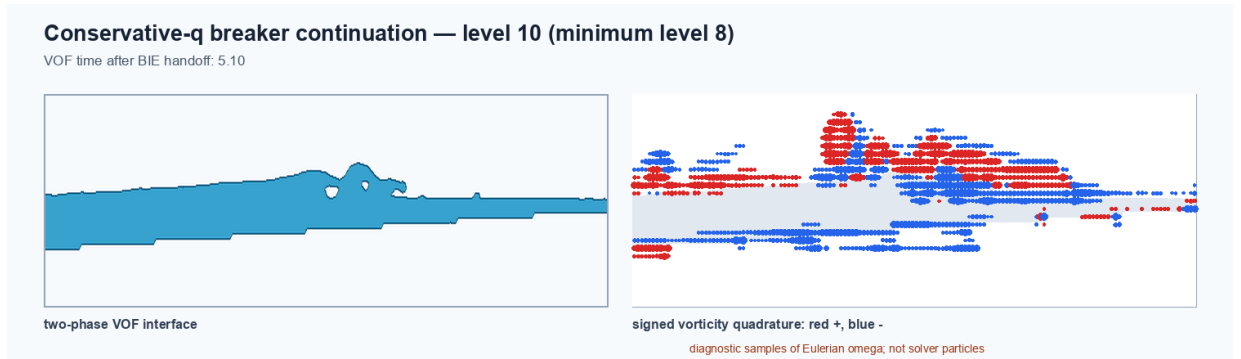


Figure 6: Level-10 state at $t = 5.10$. Left: two-phase VOF interface. Right: signed samples of the resolved Eulerian vorticity (red positive, blue negative).

5 Discussion

The experiment separates stability from physical convergence. Transporting q with shared embedded fluxes removes the previous long-time mass loss, and the receiver-grid correction retains divergence and momentum compatibility after AMR. These properties remain stable through 301 output times on all three grids. They do not, however, determine the topology of the free surface: the first resolved gas component appears only on level 10 and is transient, while the upper-surface geometry remains outside the selected fine-grid tolerance.

The level-10 cavity may be the first resolved representation of jet closure, or it may be a resolution-dependent branch selected by the coupled AMR and interface reconstruction. The current data cannot distinguish these possibilities. A decisive impact calculation requires at least one finer receiver grid, time-step refinement, several earlier handoff times and an aligned experimental or high-resolution two-phase reference. Event time, jet-tip position, enclosed-gas area and pressure impulse should converge together. Air compressibility and three-dimensionality may also become important near closure.

The contribution is therefore the conservative and divergence-conforming coupling, together with a three-level observation that topology converges more slowly than invariants. A matched end-to-end accuracy–cost comparison with a contemporary BEM–VOF or potential-flow/particle hybrid remains necessary to establish an efficiency advantage. The present GPU timing concerns only the sparse handoff reconstruction; the serial two-phase solve dominates the reported continuation cost.

6 Conclusions

An adaptive Euler–BIE state can be transferred to an embedded two-phase VOF receiver without sacrificing the receiver’s discrete divergence structure. The essential choices are to transfer shared face averages, construct the MAC field as a streamfunction curl, impose momentum in the receiver quadrature and transport the full-cell liquid aperture $q = fc_s$ with a shared embedded-boundary flux.

Manufactured tests show first-order aperture convergence with roundoff mass and bound preservation. The coupled L8–L10 breaker remains bounded through $t = 6$, with maximum relative liquid-volume drift below 8.57×10^{-8} and small post-AMR compatibility corrections. The level-10 trajectory contains a resolved but transient gas component and strong crest vorticity. Its onset and morphology are not reproduced on the coarser grids, and the final L9/L10 profile and crest differences exceed 5%.

The present result supports a conservative hybrid continuation method and identifies a resolution-dependent topology transition. It does not yet support a grid-converged prediction of plunging impact, trapped-air volume or impact impulse.

Data and code availability

Source code, solver inputs, machine-readable diagnostics and scripts used to produce the figures are available in the accompanying research repository. The BIE input data and external comparison data retain their original licences. The two-phase calculations use a pinned Basilisk environment; the exact revision and compile parameters are supplied with the archived runs.

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